Seminar Task 3: Mini Data Science Project - From Question to Insight

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Predictive Maintenance in Industrial Manufacturing

1. **Problem Formulation:**

In the manufacturing sector, unplanned machine breakdown is a major problem and due to that production modules get idle and disrupt supply chains. As the end result, the operational cost will get higher. The traditional machine maintenance method is repair the machine after a breakdown or maintain the machine for a rigid schedule. If we can identify a machine that has a possible risk breakdown before it happens, we can optimize the unplanned machine downtime and optimize the machine maintenance.

The Research Question: Can machine learning architectures accurately predict industrial equipment failure using real-time sensor metrics before a breakdown occurs?

.Throughout this mini project, several machine learning models will be trained and evaluated using machine sensor data to build an accurate predictive model.

2. **Data Acquisition:**

The AI4I 2020 Predictive Maintenance Dataset is a publicly available synthetic dataset hosted on the UCI Machine Learning Repository, designed to reflect real predictive maintenance scenarios encountered in industry. It has been widely used in industrial AI benchmarks.

Source: https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset

The dataset contains 10,000 rows, each representing a single machine observation, with 14 columns:

Feature	Type	Description
Air temperature [K]	Continuous	Ambient air temperature (sensor reading, range 295–304 K)
Process temperature [K]	Continuous	Temperature at the machining process (range 306–314 K)
Rotational speed [rpm]	Continuous	Spindle/tool rotation speed
Torque [Nm]	Continuous	Applied torque on the machine tool
Tool wear [min]	Continuous	Cumulative time the tool has been in use
Product Type (H/M/L)	Categorical	Product quality variant: High, Medium, Low
Machine failure	Binary	Target: 1 if any failure occurred, 0 otherwise
TWF / HDF / PWF / OSF / RNF	Binary	Specific failure type indicators

3. **Data Analysis & AI Techniques:**

3.1. Data Cleaning & Preprocessing

Within the dataset, no missing values were detected across all 14 columns and the dataset is clean. There is no requirement to do imputation. Machine failure records account for 3.39% of the total records, which is consistent with the real-world scenario.

Two engineered features were derived from the existing data:

- Temperature Differential = Process Temp − Air Temp, which captures heat build-up
- Mechanical Power = Torque × Rotational Speed × (2π/60), representing actual energy input to the machine.

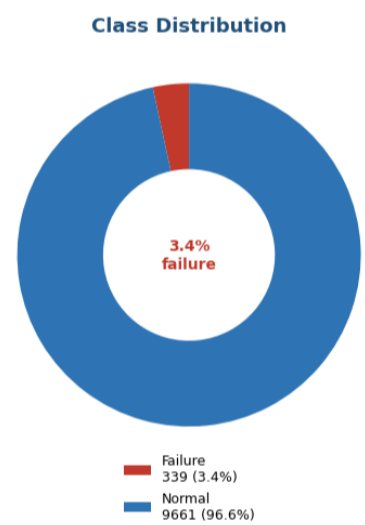
To ensure equal contribution to distance-based computations, StandardScaler normalization was applied to all features before model training.

Class imbalance was addressed using class_weight="balanced" in Random Forest and Logistic Regression, and via Stratified K-Fold cross-validation.

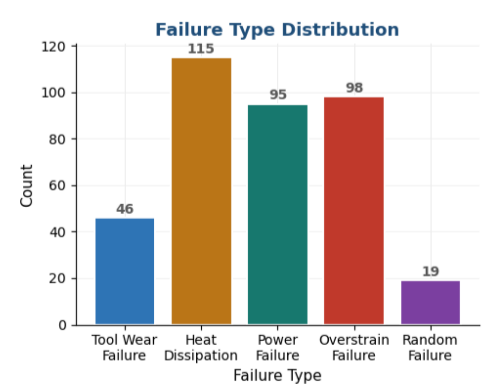
3.2. Exploratory Data Analysis

Key observations from EDA:

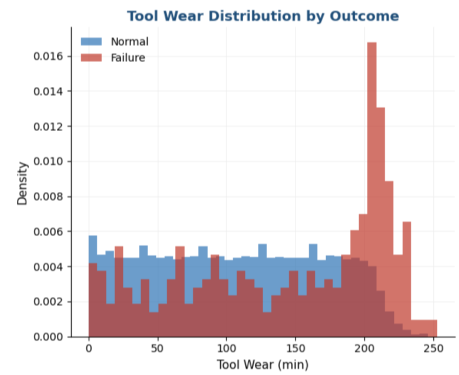
- Class imbalance is severe — only 339 failures out of 10,000 records (3.39%). Standard accuracy metrics are misleading; AUC-ROC and Average Precision are more appropriate.



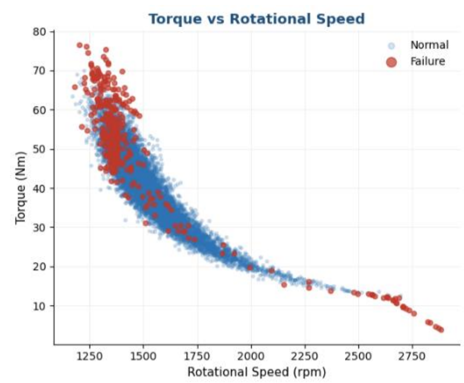
- Heat dissipation failure (HDF, n=115) is the most common failure type, followed by overstrain (OSF, n=98) and power failure (PWF, n=95). Random failure (RNF, n=19) is the rarest.



- Failed machines tend to have higher tool wear values, confirming that cumulative tool degradation is a key risk factor.

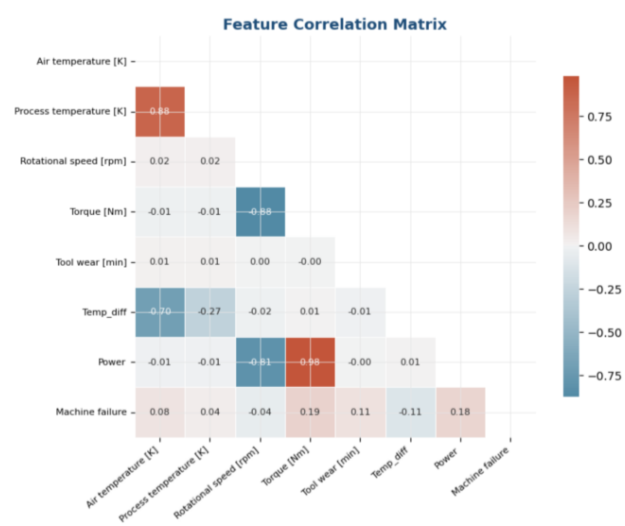


- The torque-speed scatter plot reveals a distinct operating envelope. Failures cluster at the extremes — high torque combined with low or very high rotational speed — suggesting mechanical stress as a root cause



Feature Analysis:

- Torque and rotational speed are strongly negatively correlated (−0.88), which is physically expected: machines apply higher torque at lower speeds.
- Process temperature and air temperature are highly correlated (0.88), making the derived temperature differential a more informative feature.
- Tool wear has a modest positive correlation with machine failure (0.10), but its importance is amplified in nonlinear models.



3.3. Machine Learning Methods

Three machine learning architectures were implemented to compare baseline linear limits against non-linear tree ensembles:

- Logistic Regression (Baseline): Utilized with class weight='balanced' to establish a linear performance floor.
- Random Forest Classifier: An ensemble method deployed to handle non-linear sensor interactions (e.g., high torque paired with high tool wear).
- XGBoost Classifier: A gradient boosting architecture utilizing scale_pos_weight to aggressively penalize the model for missing minority-class breakdown events.

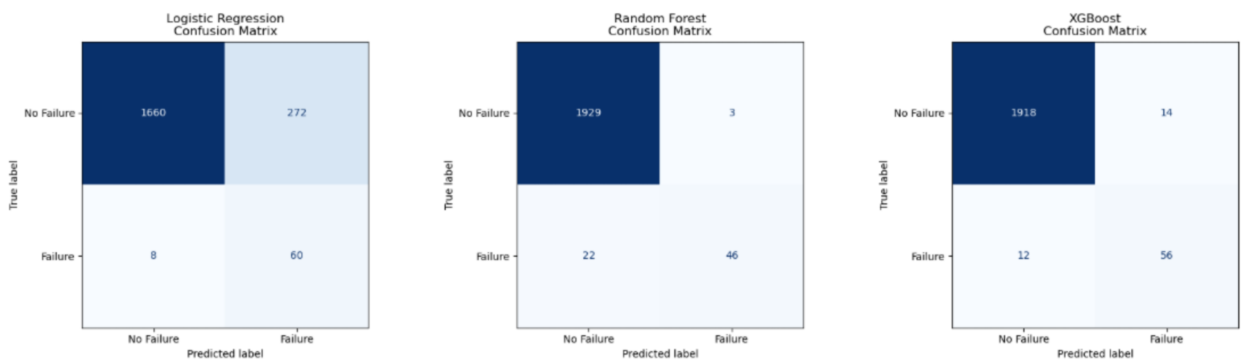
4. Results & Interpretation:

4.1. Performance comparison

The following performance metrics were obtained from the comparative analysis of the stratified test set

Model	AUC-ROC	Avg Precision	Failure Recall	Failure F1
Logistic Regression	0.9330	0.4733	0.8824	0.3000
Random Forest	0.9674	0.8515	0.6765	0.7863
XGBoost	0.9780	0.8663	0.8235	0.8116

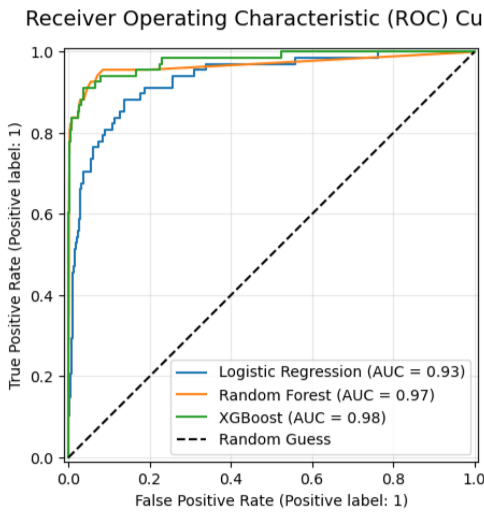
4.2. Confusion matrices

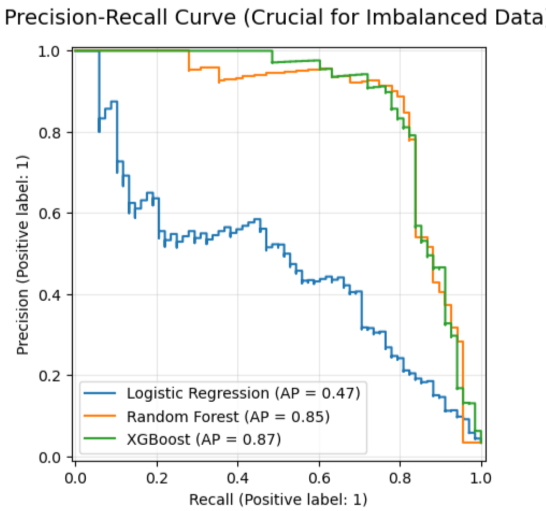


4.3. Key insights of the results

- Logistic Regression classifier catches most failures (88% Recall) but triggers massive false alarms (18% Precision), making it operationally useless due to alarm fatigue.
- As in the results, the Random Forest classifier eliminates false alarms (94% Precision) but misses nearly a third of actual breakdowns (67% Recall), leaving the plant highly vulnerable to unplanned downtime.
- We can say that the XGBoost Classifier is the winner here because it strikes the perfect equilibrium (~81% Precision and 82% Recall). It achieves the highest overall F1-Score and

Average Precision, making it the safest, most cost-effective choice for real-world shopfloor deployment.





4.4. Conclusion

Ultimately, this project proves that machine failure can be accurately predicted using sensor data and targeted feature engineering. The selected XGBoost classifier achieves an AUC-ROC of 0.9780, correctly ranking failing machines above healthy ones 97.8% of the time. More importantly for operational deployment, it maintains a strong Average Precision (0.8663) and F1-Score (0.8116), proving its ability to catch critical failures without overwhelming floor engineers with false alarms.

5. Reflection

5.1. Challenges Faced and Learnings

- Class imbalance was the primary challenge. With only 3.39% positive cases, a naive classifier achieves 96.6% accuracy by predicting "no failure" for every observation and zero actual failures.
  So for this challenge, rather than based on the accuracy, learned to choose the appropriate metrics like AUC-ROC, Average Precision, F1 to get the idea how well the model predicts machine failures.
- Furthermore, diagnosing the high correlation between thermal sensors taught me the value of domain-driven feature engineering; synthesizing raw inputs into mechanical performance metrics drastically improved the model's predictive separation.

Since this is a real-world problem stays in the industry I have currently working on, I have interested in this project. After the related public dataset has found from the internet I have done this independently through personal research, coding and data analysis.